Update the weights by moving in the negative direction of the Gradient of the Loss function with respect to model weights

$$Net_j = \sum_i (w_{ji}\, O_i) \quad \rightarrow \quad \frac{\partial\, Net_j}{\partial\, w_{ji}} = O_i$$

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$ (For Sigmoid Function)

$$E = \frac{1}{2}\sum_j (t_j - O_j)^2$$ (Error at Output layer ONLY)

$$\Delta w_{ji} = -\eta \frac{\partial\, E}{\partial\, w_{ji}} = -\eta \frac{\partial\, E}{\partial\, Net_j} \frac{\partial\, Net_j}{\partial\, w_{ji}}$$

$$\Delta w_{ji} = -\eta \frac{\partial\, E}{\partial\, Net_j}\, O_i$$ Let $$\delta_j = -\frac{\partial\, E}{\partial\, Net_j}$$

$$\Delta w_{ji} = \eta\, \delta_j\, O_i$$ then $$w_{ji}^{new} = w_{ji}^{old} + \Delta w_{ji}$$